

Linear Independence as a Linear Combination

The following theorem provides a familiar way to check if a set of vectors is linearly independent.

Theorem

The collection of vectors, $\{\vec{u}_1, \dots, \vec{u}_k\}$ in $\mathbb{R}^n$ is linearly independent if and only if whenever

$$\sum_{i=1}^n a_i \vec{u}_i = \vec{0}$$

it follows that each $a_i = 0$. Thus $\{\vec{u}_1, \dots, \vec{u}_k\}$ in $\mathbb{R}^n$ is linearly independent exactly when the system of linear equations $AX = 0$ has only the trivial solution, where A is the $n \times k$ matrix having these vectors as columns.

Problem

Let

$$A = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 2 & -1 & 0 & 1 \\ 3 & 1 & 3 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Express B as a linear combination of the columns A_1, A_2, A_3, A_4 of A , or show that this is impossible.

Consider $C = [A \ B]$. As stated in the Theorem above, if $C\vec{X} = \vec{0}$ has only the trivial solution, then $\{\vec{A}_1, \vec{A}_2, \vec{A}_3, \vec{A}_4, \vec{B}\}$ is a linearly independent set. If this is the case, we will not be able to write $\vec{B}$ as a linear combination of columns of A . Also, if $C\vec{X} = \vec{0}$ has a nontrivial solution, then we have $a_1 \vec{A}_1 + a_2 \vec{A}_2 + a_3 \vec{A}_3 + a_4 \vec{A}_4 + a_5 \vec{B} = \vec{0}$. If $a_5 \neq 0$, then

$$\vec{B} = -\frac{1}{a_5} (a_1 \vec{A}_1 + a_2 \vec{A}_2 + a_3 \vec{A}_3 + a_4 \vec{A}_4).$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & 0 \\ 2 & -1 & 0 & 1 & 1 & 0 \\ 3 & 1 & 3 & 1 & 1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & 0 \\ 0 & -1 & -4 & 3 & -1 & 0 \\ 0 & 1 & -3 & 4 & -2 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & 0 \\ 0 & -1 & -4 & 3 & -1 & 0 \\ 0 & 0 & -7 & 7 & -3 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & -1 & 1 & 0 \\ 0 & 1 & 4 & -3 & 1 & 0 \\ 0 & 0 & 1 & -1 & \frac{3}{7} & 0 \end{array} \right]$$

$$x_1 = -2(s - \frac{3}{7}t) + s - t = -s - \frac{1}{7}t$$

$$x_2 = -4(s - \frac{3}{7}t) + 3s - t = -s + \frac{5}{7}t$$

$$x_3 = s - \frac{3}{7}t$$

$$x_4 = s$$

$$x_5 = t$$

where $s, t \in \mathbb{R}$

For $s=1$ and $t=-1$: $x_1 = -\frac{6}{7}$ $x_2 = -\frac{12}{7}$
 $x_3 = \frac{10}{7}$ $x_4 = 1$
 $x_5 = -1$

$$C\vec{X} = x_1\vec{A}_1 + x_2\vec{A}_2 + x_3\vec{A}_3 + x_4\vec{A}_4 + x_5\vec{B} = \vec{0}$$

$$-\frac{6}{7}\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \frac{12}{7}\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + \frac{10}{7}\begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} - 1\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

As an example, $\vec{B}$ can be written as the following linear combination of the columns of A .

$$\vec{B} = -\frac{6}{7}\vec{A}_1 - \frac{12}{7}\vec{A}_2 + \frac{10}{7}\vec{A}_3 + \vec{A}_4$$

Theorem: Any set $\{\vec{A}_1, \vec{A}_2, \dots, \vec{A}_k\}$ in $\mathbb{R}^n$ is linearly dependent if $k > n$.

In the example above the set $\{\vec{A}_1, \vec{A}_2, \vec{A}_3, \vec{A}_4, \vec{B}\}$ in $\mathbb{R}^3$, was shown to be a linearly dependent set. Note that the number of vectors in the set is more than the number of entries in each vector.

Definition

A **transformation** is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, sometimes written $\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m$, and is called a **transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$** . The sets $\mathbb{R}^n$ and $\mathbb{R}^m$ are known as the **domain** and **codomain** of T . If $m = n$, then we say **T is a transformation of $\mathbb{R}^n$** .

Example:

Consider $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ defined by $T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ y \\ y-z \\ y+z \end{bmatrix}$. This transformation, transforms $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \in \mathbb{R}^3$ to $\begin{bmatrix} 1 \\ 0 \\ -2 \\ 2 \end{bmatrix} \in \mathbb{R}^4$.

What do we mean by a function?

Informally, a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a rule that assigns exactly one vector of $\mathbb{R}^m$ to each vector of $\mathbb{R}^n$. We use the notation $T(\vec{x})$ to mean the transformation T applied to the vector $\vec{x}$.

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^4, T(\vec{x}) = \vec{b}, \vec{x} \in \mathbb{R}^3, \vec{b} \in \mathbb{R}^4$$

in the above example

$$\begin{array}{l} \vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \\ \vec{b} = \begin{bmatrix} x+y \\ y \\ y-z \\ y+z \end{bmatrix} \end{array}$$

Definition

If T acts by matrix multiplication of a matrix A , we call T a **matrix transformation**, and write $T_A(\vec{x}) = A\vec{x}$. The vector $T_A(\vec{x}) = A\vec{x}$ is called the **image** of $\vec{x}$ and the set of all images is called the **range** of T .

Example:

$$T_A(\vec{x}) = x \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\vec{x}} = \begin{bmatrix} x+y \\ y \\ y-z \\ y+z \end{bmatrix}$$

Definition (Equality of transformations)

Suppose $S : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are transformations. Then $S = T$ if and only if $S(\vec{x}) = T(\vec{x})$ for every $\vec{x} \in \mathbb{R}^n$.

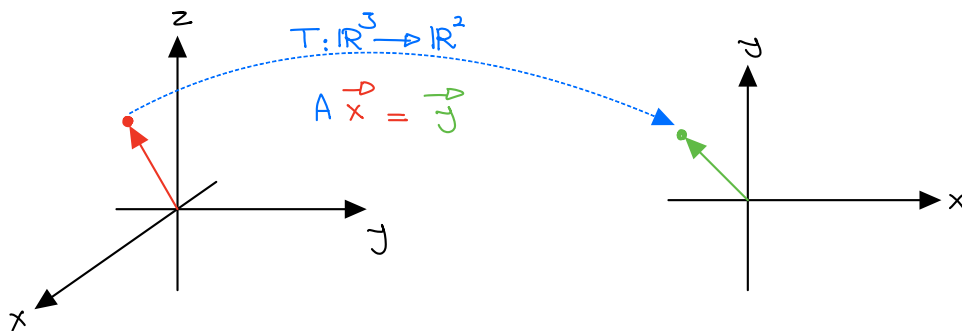
Transformation by Matrix Multiplication

Example

Consider the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix}$. By matrix multiplication, A transforms vectors in $\mathbb{R}^3$ into vectors in $\mathbb{R}^2$. Consider the vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Transforming this vector by A looks like:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix}_{2 \times 3} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1} = \begin{bmatrix} x+2y \\ 2x+y \end{bmatrix}_{2 \times 1}$$

For example: $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$



Linear Transformations

Definition

A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a **linear transformation** if it satisfies the following two properties for all $\vec{x}, \vec{y} \in \mathbb{R}^n$ and all (scalars) $a \in \mathbb{R}$.

- 1 $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$ (preservation of addition)
- 2 $T(a\vec{x}) = aT(\vec{x})$ (preservation of scalar multiplication)

Properties of Linear Transformations

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let $\vec{x} \in \mathbb{R}^n$. Since T preserves scalar multiplication,

- 1 $T(0\vec{x}) = 0T(\vec{x})$ implying $T(0) = 0$, so **T preserves the zero vector.**
- 2 $T((-1)\vec{x}) = (-1)T(\vec{x})$, implying $T(-\vec{x}) = -T(\vec{x})$, so **T preserves the negative of a vector.**

Suppose $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ are vectors in $\mathbb{R}^n$ and

$$\vec{y} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_k\vec{x}_k$$

for some $a_1, a_2, \dots, a_k \in \mathbb{R}$. Then

$$\begin{aligned} \text{3} \quad T(\vec{y}) &= T(a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_k\vec{x}_k) \\ &= a_1T(\vec{x}_1) + a_2T(\vec{x}_2) + \cdots + a_kT(\vec{x}_k), \end{aligned}$$

i.e., **T preserves linear combinations.**